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Smooth Anodic TiO₂ Nanotubes**

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In 1999 Zwilling et al. reported on the electrochemical formation of self-assembled TiO₂ nanotubes (π -TiO₂) by the anodization of Ti,^[1] and other reports followed soon thereafter.^[2,3] A factor limiting the application of this first generation of nanotubes was their production in hydrofluoric

acid based electrolytes. As a result of the high rate of chemical dissolution of TiO₂ in these solutions, the nanotubes could be grown only up to a length of 500 nm. Recently, we have shown^[4–7] and explained^[4] how a second generation of nanotubes with lengths up to several micrometers and aspect ratios up to 50 can be formed by adjusting the pH gradient within the growing nanotube. Common to all these anodic approaches is that the side walls of the tubes show strong irregularities, that is, the side-wall profiles show considerable thickness variations (ripples) as shown in Figure 1 a. In the present work we demonstrate how TiO₂ nanotubes with extremely smooth homogenous walls and aspect ratios up to 175 can be grown (as shown in Figure 1 b). This third generation of nanotubes is formed by suppressing local concentration fluctuations and pH bursts during anodization by using highly viscous glycerol electrolytes.

In previous work it has been established that the length of the nanotubes is essentially the result of a steady-state situation between electrochemical formation of TiO₂ at the pore tip and the chemical dissolution of formed TiO₂ by fluorides from the electrolyte.^[3,5] We showed how the pH at the pore tip is lowered by the hydrolysis reactions of the Ti⁴⁺ ions formed during anodization. As the Ti⁴⁺ flux can be controlled by the applied current, it was possible to adjust conditions that keep the pore tip sufficiently acidic while the pore walls dissolve much more slowly.^[4] As a result, the nanotubes could be grown up to 2.5 μ m in length, in contrast to the previous length of 500 nm (Figure 1 a). Some other reports on these micrometer-length tubes^[8,9] describe nanotubes with aspect ratios up to 44^[8] and 58.^[9] In all these reports it is apparent that relatively regular variations of the walls' thickness occur (Figure 1 a).

To explain these variations one may examine the current transients of the electrochemical anodization of Ti (Figure 2). It is apparent that relatively regular current oscillations occur during anodization, for example in sulphate electrolytes. When the frequency of the current oscillations is converted into a length scale, it compares well to the distance between ridges on the side walls. Thus, the current transients are correlated with the variations in the wall thickness. This can be explained by the fact that every current transient is accompanied by a pH burst at the pore tip. This local acidification leads to a temporarily increased dissolution rate. Since diffusion is the main effect balancing local acidification,^[4] a straightforward remedy to suppress pH bursts is to damp the fluctuations by decreasing the diffusion constant in the electrolyte. This can be achieved, for example, by using highly viscous electrolytes such as glycerol-based solutions. Clearly the current–time transient in a fluoride-containing glycerol electrolyte (Figure 2) shows much lower current densities than that in the purely aqueous electrolyte. This is in line with a diffusion-controlled process (considering the Stokes–Einstein relation, where the diffusion constant, D , is proportional to $1/\eta$, where η is the solution viscosity). But even more important than the lowering of the current by glycerol is that the curves in Figure 2 do not exhibit any detectable current fluctuations.

As a result, the tubes formed under these electrochemical conditions (Figure 1 b) are entirely smooth over their entire

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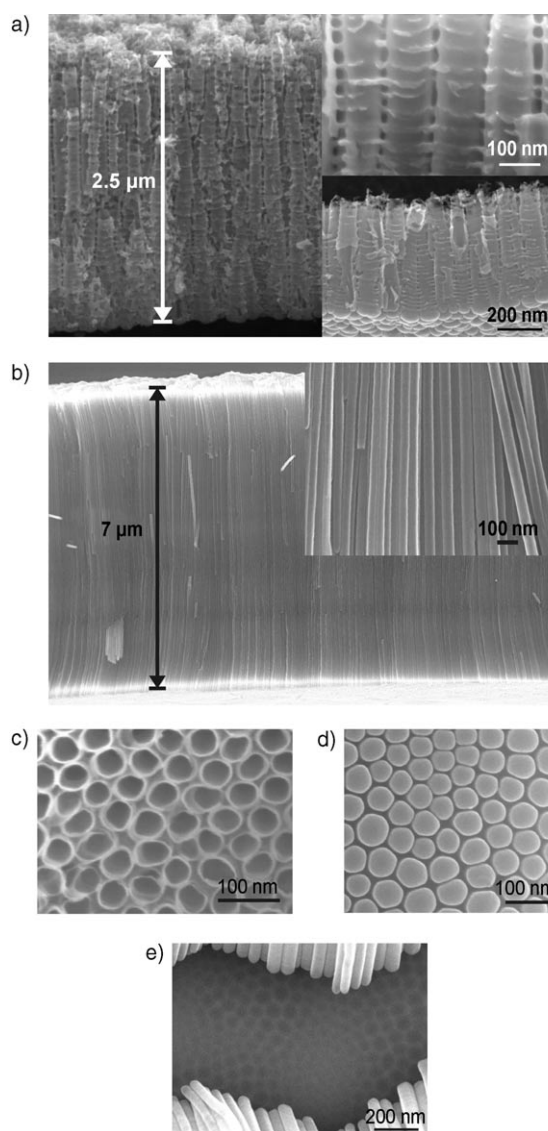


Figure 1. a) SEM images of conventional anodic TiO_2 nanotubes with marked inhomogeneities (from Ref. [4]). The insets show a magnification of the structure (top) and an example of ripples on short tubes formed in 1 M H_2SO_4 with 0.15 wt% HF (bottom). b) Smooth TiO_2 nanotubes 7- μm long produced in glycerol electrolyte with 0.5 wt% NH_4F . The inset shows the walls of the nanotubes in more detail. c)–e) SEM images of smooth TiO_2 nanotubes 7- μm long produced in glycerol electrolyte: c) top view; d) bottom view. The average nanotube diameter is approximately 40 nm and the average pore spacing is approximately 60 nm. e) Top view of an anodized sample after removal of some arrays of nanotubes. The lower part of the tubes and part of the Ti substrate surface are visible. Clearly the convex shape of the nanotube bottoms is reflected in the surface of the Ti substrate.

length. Additionally as a result of the change in the diffusion coefficients and the improved confinement of the acidic area at the pore tip, the tubes grow now to a length of 7 μm . The tubes are arranged in very tight bundles and are not connected with another; this is in contrast to all previous descriptions (Figure 1a). Figure 1c and d show scanning electron micrographic (SEM) images of self-assembled TiO_2 nanotubes from the top and bottom, respectively. It is apparent that the tubes are open on the top and closed with

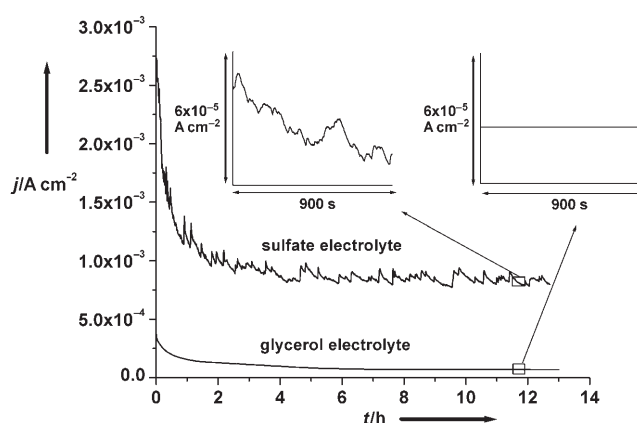


Figure 2. Current–time behavior during anodization of Ti samples at 20 V for 13 h. The samples were anodized in glycerol or in 1 M $(\text{NH}_4)_2\text{SO}_4$, in both cases with 0.5 wt% NH_4F . The insets show details of the current transients. Current fluctuations can be observed for the anodization in aqueous solution, while they are completely suppressed using the glycerol electrolyte.

a TiO_2 layer on the bottom of the layer of nanotubes. The inner diameters of the nanotubes are mostly around 40 nm; however, some tubes with diameters in the range between 30 and 60 nm can be found. In other words, tubes with aspect ratios between 233 and 117 are obtained. Figure 1e shows an SEM image of the lower part of the tubes and the interface between the TiO_2 nanotubes and the underlying Ti substrate. This image reveals that the convex shape of the closed ends of the nanotubes is reflected in the morphology of the surface of the Ti substrate. We reached a similar conclusion in our earlier work on the formation of nanotubular structures by anodization in aqueous fluoride-containing electrolytes.^[4,5] We believe that the interface of the convex (nanotubes) and concave (Ti substrate) units plays a significant role for the good adhesion of both adjacent parts.

Figure 3 shows a transmission electron microscopic (TEM) image of several nanotubes (a), the TEM diffraction pattern (b), and a high-resolution TEM image of the single nanotube bottom (c). These TEM images confirm that the walls of the nanotubes are very smooth and straight over their entire length. The thickness of the walls is 12 nm, the thickness of the convex barrier oxide is 16 nm. If TEM

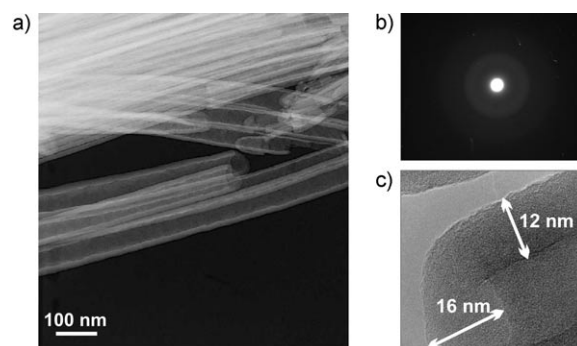


Figure 3. Smooth TiO_2 nanotubes produced in the glycerol electrolyte: a) TEM image of several TiO_2 nanotubes; b) TEM diffraction pattern of a nanotube wall; c) HRTEM image of a single nanotube.

pictures from the side wall close to the pore bottom are compared to pictures close to the pore mouth, only minimal variations in thickness, (12 ± 2) nm, can be determined. This is also a clear improvement over previous syntheses, in which considerable thinning of the walls at the pore mouth was typically observed. The TEM diffraction pattern in Figure 3b was taken from the nanotube wall and shows clearly that the TiO_2 structure is amorphous. The high-resolution TEM image in Figure 3c additionally confirms the amorphous structure in the side wall and at the bottom of the tubes.

In summary, TiO_2 nanotubes prepared in glycerol solutions show an unprecedented degree of regularity and homogeneity in comparison to the nanotubes described in earlier reports, in which thinning of the tube wall towards the mouth and ripples on the tube walls were observed.^[1–9] In addition, the pore diameter is strongly affected by the electrolyte: in aqueous electrolytes it is about 100 nm, in glycerol it is about 40 nm. Besides the high aspect ratio, the smoothness of the nanotubes plays a crucial role in the improvement of various properties of the layers of self-assembled TiO_2 nanotubes. For instance, uniform depth profiles and flatter interfaces are essential from the point of view of optical devices, for example, waveguides and photonic crystals, where the light transmission is severely reduced by diffuse scattering at morphological inhomogeneities. Because of the high refractive index of TiO_2 (2.6), self-assembled nanotubular TiO_2 structures are viable candidates for applications in photonic crystals^[10] and waveguides.^[11]

We also note that layers of nanotubes with a tube length of 7 μm may be suitable for application in highly efficient dye-sensitized solar cells,^[12] since the desired light absorption depth, nanoparticulate TiO_2 films with a thickness in the range of 5 to 10 μm yield optimum efficiency. In addition to dye sensitization,^[13] several other applications have already been reported for the self-assembled TiO_2 nanotubes, for example, as a catalyst support for Pt/Ru nanoparticles for the electrooxidation of methanol^[14] and as structures that can be modified with self-assembled monolayers of organic molecules to tailor surface wettability.^[15]

Other potential applications of self-assembled TiO_2 nanotubular structures may be based on exploiting the specific properties of TiO_2 , such as its catalytic properties,^[16] self-cleaning,^[17] and semiconductor properties.^[18] The nanotubular structure could be also used to improve biocompatibility in biomedical applications.^[19] Our results provide greatly improved nanotubular TiO_2 substrates for all these applications and represent a significant advancement towards the “ideal” tubular shape.

Experimental Section

Surface treatment: Ti samples (0.1-mm thick foils, 99.6% purity, Goodfellow, England) were degreased by sonicating in acetone, 2-propanol, and methanol, followed by rinsing with deionized water and drying in a nitrogen stream.

Electrochemical treatment: Titanium samples were contacted with a Cu plate and then pressed against an O-ring in an electrochemical cell, such that 1 cm^2 was exposed to the electrolyte. The setup consisted of a conventional three-electrode configuration with a platinum gauze as a counterelectrode and a Haber–Luggin capillary

with Ag/AgCl (1M KCl) electrode as a reference electrode. Electrochemical experiments were carried out using a high-voltage potentiostat Jaissle IMP 88. Electrolytes were 0.5 wt % NH_4F in glycerol (1,2,3-propanetriol) with a dynamic viscosity of 1350 mPa s at 20 °C or in 1M $(\text{NH}_4)_2\text{SO}_4$. The electrolytes were prepared from reagent-grade chemicals. The electrochemical treatment consisted of a potential ramp from the open-circuit potential (OCP) to 20 V with a sweep rate of 1 V s^{-1} , after which the applied potential was held at 20 V for 13 h. The same treatment was used for reference samples produced in aqueous electrolytes. An estimate of the IR drop during experiments in glycerol electrolyte results in an error of less than 1 V on the effective potential. After the electrochemical treatment the samples were soaked with deionized water and dried in a nitrogen stream.

Surface characterization: Scanning electron microscopy (Hitachi FE-SEM S4800) was employed for the morphological characterization of the TiO_2 nanotubes. To gain information on thickness of the nanotubular layer, direct SEM cross-sectional thickness measurements were carried out on mechanically bent and cracked samples. This technique was also carried out to achieve bottom-view images of the nanotube layers. Transmission electron microscope Philips CM 30 T/STEM and high-resolution TEM Philips CM 300 UT were used for the morphological and structural characterization of the TiO_2 nanotubes.

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